

PORTFOLIO 5: STRINGS

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Multi-Code String Encoder System (Morse, Caesar Cipher, and Binary Translator)

This portfolio features a Multi-Code String Encoder System that allows users to convert text into different encoded formats such as Morse Code, Caesar Cipher, and Binary Code, as well as decode them back into readable text. The project demonstrates how string manipulation can be applied across multiple encoding systems using core programming concepts including Strings, Functions, Conditionals, and Iterations to build a structured and interactive translation tool.

Key techniques include string normalization using methods like `.upper()` and `.strip()` to standardize user input, definite loops to process each character in a string, and string construction and joining using concatenation (+) and `" ".join()` to build encoded outputs. The system applies multi-way conditionals (if, elif) to determine which encoding method is used, while also using ASCII conversion functions like `ord()` and `chr()` for Caesar Cipher and Binary translation. Additionally, a while loop provides continuous user interaction, and input validation ensures the program runs smoothly across different encoding modes and decoding operations.

Morse Code Dictionary

Stores the characters and their Morse code equivalents for translation.

```
In [1]: MORSE_DICT = {
    'A': '.-.', 'B': '-...', 'C': '-.-.', 'D': '-..', 'E': '.',
    'F': '..-.', 'G': '--.', 'H': '....', 'I': '..', 'J': '-.-.-',
    'K': '-.-', 'L': '..-..', 'M': '--', 'N': '-.', 'O': '---',
    'P': '-.-.', 'Q': '-.-.-', 'R': '.-..', 'S': '...',
    'T': '-.', 'U': '..-.', 'V': '...-', 'W': '.--',
    'X': '-.-.-', 'Y': '-.-.-', 'Z': '-.-.-',
    '0': '-----', '1': '.-----', '2': '..---', '3': '...--',
    '4': '....-', '5': '.....', '6': '-.....', '7': '-.-...',
    '8': '-.-.-.', '9': '-.-.-.-', ' ': '/'
}
```

Reverse Morse Dictionary

Creates a reverse lookup so Morse code can be converted back into text.

```
In [2]: REVERSE_MORSE = {v: k for k, v in MORSE_DICT.items()}
```

Morse Code Functions

Converts text to Morse code and Morse code back to text using string processing.

```
In [3]: def text_to_morse(text):
        text = text.upper().strip()
        return " ".join(MORSE_DICT.get(c, "") for c in text).strip()

def morse_to_text(code):
    words = code.strip().split(" ")
    return "".join(REVERSE_MORSE.get(c, "") for c in words)
```

Caesar Cipher Functions

Shifts letters forward or backward in the alphabet using ASCII values.

```
In [4]: def caesar_encrypt(text, shift):
        text = text.upper().strip()
        result = ""

        for char in text:
            if char.isalpha():
                result += chr(((ord(char) - 65 + shift) % 26) + 65)
            else:
                result += char

        return result

def caesar_decrypt(text, shift):
    return caesar_encrypt(text, -shift)
```

Binary Conversion Functions

Converts text into 8-bit binary and back using ASCII values.

```
In [5]: def text_to_binary(text):
        text = text.upper().strip()
        return " ".join(format(ord(c), "08b") for c in text)

def binary_to_text(binary):
    return "".join(chr(int(b, 2)) for b in binary.split())
```

Program Menu

Displays all available tools clearly so the user understands what each option does.

```
In [6]: print("==== STRING MULTI-ENCODER SYSTEM =====")
        print("1 - Text ↔ Morse Code")
        print("2 - Text ↔ Caesar Cipher")
        print("3 - Text ↔ Binary Code")
```

```

print("4 - Exit")

print("\nNOTE:")
print("- Morse uses '.' and '-'")
print("- Caesar uses shift (e.g., 3 means A → D)")
print("- Binary uses 8-bit ASCII codes")

```

===== STRING MULTI-ENCODER SYSTEM =====

- 1 - Text ↔ Morse Code
- 2 - Text ↔ Caesar Cipher
- 3 - Text ↔ Binary Code
- 4 - Exit

NOTE:

- Morse uses '.' and '-'
- Caesar uses shift (e.g., 3 means A → D)
- Binary uses 8-bit ASCII codes

Main Program Loop

Controls user input, mode selection, and runs all encoding/decoding functions.

```

In [11]: while True:
            print("\n===== STRING ENCODER SYSTEM =====")
            print("1 - Morse Code")
            print("2 - Caesar Cipher")
            print("3 - Binary Code")
            print("4 - Exit")

            choice = input("\nSelect option (1-4): ")

            if choice == "4":
                print("Program ended.")
                break

            # FIRST: ask mode (better UX)
            mode = input("Choose mode (e = encode / d = decode): ")

            # THEN: input text
            text = input("Enter input: ")

            # ----- MORSE -----
            if choice == "1":
                if mode == "e":
                    print("Morse:", text_to_morse(text))
                else:
                    print("Text:", morse_to_text(text))

            # ----- CAESAR -----
            elif choice == "2":
                shift = int(input("Enter shift value: "))

                if mode == "e":
                    print("Encrypted:", caesar_encrypt(text, shift))
                else:
                    print("Decrypted:", caesar_decrypt(text, shift))

            # ----- BINARY -----
            elif choice == "3":
                if mode == "e":

```

```

        print("Binary:", text_to_binary(text))
    else:
        print("Text:", binary_to_text(text))

    else:
        print("Invalid option. Please choose 1-4.")

```

===== STRING ENCODER SYSTEM =====

- 1 - Morse Code
- 2 - Caesar Cipher
- 3 - Binary Code
- 4 - Exit

Morse: -... .- -.- .- ... -.- - - / - . - / .- .- .- .- .- .

===== STRING ENCODER SYSTEM =====

- 1 - Morse Code
- 2 - Caesar Cipher
- 3 - Binary Code
- 4 - Exit

Text: BAKASYON NA PLEASE

===== STRING ENCODER SYSTEM =====

- 1 - Morse Code
- 2 - Caesar Cipher
- 3 - Binary Code
- 4 - Exit

Decrypted: 1

===== STRING ENCODER SYSTEM =====

- 1 - Morse Code
- 2 - Caesar Cipher
- 3 - Binary Code
- 4 - Exit

Binary: 01001000 01000101 01001100 01010101 00100000 01001010 01000001 01001101
00100000 01001001 01001101 01001110 01001001 01000100 01000001

===== STRING ENCODER SYSTEM =====

- 1 - Morse Code
- 2 - Caesar Cipher
- 3 - Binary Code
- 4 - Exit

Program ended.